In [1]:
```
import numpy as np
import pandas as pd

from scipy.stats import norm
```

Liam Hopkins Problem Set 5 Questions 4-7

Generative AI Use:

Generative AI (Claude, Gemini) was used to assist in completing this assignment. AI was primarily used to explain course concepts, explain different functions and what needs to be passed to them, and explain warnings and error messages.

Question 4

For `./data/ames_prices.csv`, define an indicator equal to 1 when `Central.Air == 'Y'`. Estimate the proportion of homes with central air and test $H_0 : p = .90$. Treat a proportion as the mean of a binary variable and complete the full inference routine.

In [2]:
```
prices = pd.read_csv('ames_prices.csv')

# Setting 'Y' = 1 and 'N' = 0
prices['Central.Air'] = prices['Central.Air'].replace("Y", 1)
prices['Central.Air'] = prices['Central.Air'].replace("N", 0)
```

```
C:\Users\lchop\AppData\Local\Temp\ipykernel_31292\2513531749.py:5: FutureWarning: Do
wncasting behavior in `replace` is deprecated and will be removed in a future versio
n. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. T
o opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting',
True)`
  prices['Central.Air'] = prices['Central.Air'].replace("N", 0)
```

In [3]:
```
estimate = prices['Central.Air'].mean() # estimate is mean of our sample in this ca
print(f"estimated proportion = {estimate}")
```

```
estimated proportion = 0.9331058020477816
```

We estimate that the proportion of homes with central air is about 0.933.

Bootstrapping

In [4]:
```
## Bootstrapping:
R = 2_000

np.random.seed(5030)    # Setting the random seed

sample_props = []

for t in range(R):
```

```
    # Generating the sample
    idx = np.random.randint(0, len(prices), len(prices))
    sample = prices.iloc[idx]

    # Proportion of 'Yes'
    o_prop = sample['Central.Air'].mean()

    sample_props.append(o_prop)
```

Estimating $\hat{SE}(\hat{\theta})$

In [5]:
```
standard_error = np.std(sample_props, ddof=1)
print(f'standard error = {standard_error:.5f}')
```

```
standard error = 0.00449
```

Constructing 90%, 95%, 99% CIs

In [6]:
```
# For 90%
lower_90 = estimate - 1.645 * standard_error
upper_90 = estimate + 1.645 * standard_error

# For 95%
lower_95 = estimate - 1.960 * standard_error
upper_95 = estimate + 1.960 * standard_error

# For 99%
lower_99 = estimate - 2.576 * standard_error
upper_99 = estimate + 2.576 * standard_error

print(f"90% CI: [{lower_90:.4f}, {upper_90:.4f}]")
print(f"95% CI: [{lower_95:.4f}, {upper_95:.4f}]")
print(f"99% CI: [{lower_99:.4f}, {upper_99:.4f}]")
```

```
90% CI: [0.9257, 0.9405]
95% CI: [0.9243, 0.9419]
99% CI: [0.9215, 0.9447]
```

Testing Hypotheses

In [7]:
```
z = (estimate - 0.9)/standard_error
p = 2 * norm.cdf(-np.abs(z))

print(f"z-statistic: {z:.4f}, p-value: {p}")
```

```
z-statistic: 7.3734, p-value: 1.6629909790575168e-13
```

With the calculated z-statistic, we get a p-value well below all three significance levels of $\alpha = 0.10, 0.05, 0.01$. This means at all three significance levels we reject the null hypothesis that $\mu = 0.9$.

This decision is confirmed by our confidence intervals. 0.9 does not fall in any of the three confidence intervals, meaning these intervals also tell us that we reject the null hypothesis at

$\alpha = 0.10, 0.05, 0.01.$

Interpretation

Our estimate approximates that the proportion of homes that have central air is about 0.933 with about 0.00449 standard error. Our confidence intervals tell us that we cannot reject mean values from $[0.9257, 0.9405]$ at the $\alpha = 0.10$ level, values from $[0.9243, 0.9419]$ at the $\alpha = 0.05$ level, and values from $[0.9215, 0.9447]$ at the $\alpha = 0.01$ level. If we calculated these intervals many times, approximately 90%/95%/99% of them would contain fixed, unknown proportion of homes with central air. Our hypothesis test tells us that we reject the null hypothesis and conclude that the proportion of homes with central air is not equal to 0.9.

Question 5

For `./data/ames_prices.csv`, estimate the Pearson correlation between `arcsinh(area)` and `arcsinh(price)`. Bootstrap paired rows, construct all three CIs, and test $H_0 : \rho = 0$. Explain what would go wrong if area and price were resampled separately.

In [8]:
```
prices['arcsin_area'] = np.arcsinh(prices['area'])
prices['arcsin_price'] = np.arcsinh(prices['price'])

estimate = prices[['arcsin_area', 'arcsin_price']].corr().iloc[0, 1]
print(f"Pearson Correlation = {estimate:.4f}")
```

```
Pearson Correlation = 0.7232
```

We estimate a Pearson correlation of 0.7232 between `arcsinh(area)` and `arcsinh(price)`. The two appear to have strong positive correlation, as the value is relatively close to 1.

Bootstrapping

In [9]:
```
## Bootstrapping:
R = 2_000

np.random.seed(6001)    # Setting the random seed

sample_corrs = []

for t in range(R):
    # Generating the sample
    idx = np.random.randint(0, len(prices), len(prices))
    sample = prices.iloc[idx]

    # Correlation between the two features
    o_corr = sample[['arcsin_area', 'arcsin_price']].corr().iloc[0, 1]

    sample_corrs.append(o_corr)
```

Estimating $SE(\hat{\theta})$

In [10]:
```
standard_error = np.std(sample_corrs, ddof=1)
print(f'standard error = {standard_error:.5f}')
```

```
standard error = 0.00994
```

Constructing 90%, 95%, 99% CIs

In [11]:
```
# For 90%
lower_90 = estimate - 1.645 * standard_error
upper_90 = estimate + 1.645 * standard_error

# For 95%
lower_95 = estimate - 1.960 * standard_error
upper_95 = estimate + 1.960 * standard_error

# For 99%
lower_99 = estimate - 2.576 * standard_error
upper_99 = estimate + 2.576 * standard_error

print(f"90% CI: [{lower_90:.4f}, {upper_90:.4f}]")
print(f"95% CI: [{lower_95:.4f}, {upper_95:.4f}]")
print(f"99% CI: [{lower_99:.4f}, {upper_99:.4f}]")
```

```
90% CI: [0.7068, 0.7395]
95% CI: [0.7037, 0.7426]
99% CI: [0.6976, 0.7488]
```

Testing Hypotheses

In [12]:
```
z = estimate/standard_error
p = 2 * norm.cdf(-np.abs(z))

print(f"z-statistic: {z:.4f}, p-value: {p}")
```

```
z-statistic: 72.7788, p-value: 0.0
```

With the calculated z-statistic, we get a p-value well below all three significance levels of $\alpha = 0.10, 0.05, 0.01$. This means at all three significance levels we reject the null hypothesis that $\rho = 0$.

This decision is confirmed by our confidence intervals. 0 does not fall in any of the three confidence intervals, meaning these intervals also tell us that we reject the null hypothesis at $\alpha = 0.10, 0.05, 0.01$.

Interpretation

Our estimate approximates that the Pearson correlation between `arcsinh(area)` and `arcsinh(price)` is about 0.723 with about 0.00994 standard error. Our confidence intervals tell us that we cannot reject correlation values from $[0.7068, 0.7395]$ at the

$\alpha = 0.10$ level, values from $[0.7037, 0.7426]$ at the $\alpha = 0.05$ level, and values from $[0.6976, 0.7488]$ at the $\alpha = 0.01$ level. If we calculated these intervals many times, approximately 90%/95%/99% of them would contain fixed, unknown correlation between `arcsinh(area)` and `arcsinh(price)`. Our test hypothesis test tells us that we reject the null hypothesis and conclude that the correlation between `arcsinh(area)` and `arcsinh(price)` is not equal to 0.

If area and price were resampled seperately, this could be a problem because if they are highly correlated, area values are paired with price values. A house with a certain area is associated with a certain price. Seperately resampling them splits up these pairs, putting each area with a price it is not associated with. This could lead to the variables being found to not be highly correlated when they actually are.

Question 6

For `./data/metabric.csv`, use `Overall Survival (Months)` to estimate the mean recorded time to death or last follow-up, including both living and deceased patients. Let μ denote the population mean of this recorded-time variable, and test $H_0 : \mu = 120$ months. Complete the full inference routine and interpret the CIs in both months and years. Living patients' eventual death times are right-censored, so this exercise does not estimate unrestricted mean lifetime. Explain that distinction in your interpretation.

In [13]:
```
metabric = pd.read_csv('metabric.csv')

estimate = metabric['Overall Survival (Months)'].mean()
print(f"Mean time to death (months) = {estimate}")
print(f"Mean time to death (years) = {estimate / 12}")
```

```
Mean time to death (months) = 129.09429138744105
Mean time to death (years) = 10.757857615620088
```

We estimate the mean time to death, including both living and deceased patients, is about 129 months or about 11 years.

Bootstrapping

In [14]:
```
## Bootstrapping:
R = 2_000

np.random.seed(6021)    # Setting the random seed

sample_months = []
sample_years = []

for t in range(R):
    # Generating the sample
    idx = np.random.randint(0, len(metabric), len(metabric))
    sample = metabric.iloc[idx]
```

```
    # Correlation between the two features
    o_months = sample['Overall Survival (Months)'].mean()

    sample_months.append(o_months)
    sample_years.append(o_months / 12)
```

Estimating $SE(\hat{\theta})$

In [15]:
```
standard_error_months = np.std(sample_months, ddof=1)
standard_error_years = np.std(sample_years, ddof=1)
print(f'standard error months = {standard_error_months:.5f}')
print(f'standard error years = {standard_error_years:.5f}')
```

```
standard error months = 2.16791
standard error years = 0.18066
```

Constructing 90%, 95%, 99% CIs

In [16]:
```
estimate_years = estimate / 12

# For 90%
lower_90_months = estimate - 1.645 * standard_error_months
upper_90_months = estimate + 1.645 * standard_error_months
lower_90_years = estimate_years - 1.645 * standard_error_years
upper_90_years = estimate_years + 1.645 * standard_error_years

# For 95%
lower_95_months = estimate - 1.960 * standard_error_months
upper_95_months = estimate + 1.960 * standard_error_months
lower_95_years = estimate_years - 1.960 * standard_error_years
upper_95_years = estimate_years + 1.960 * standard_error_years

# For 99%
lower_99_months = estimate - 2.576 * standard_error_months
upper_99_months = estimate + 2.576 * standard_error_months
lower_99_years = estimate_years - 2.576 * standard_error_years
upper_99_years = estimate_years + 2.576 * standard_error_years

print(f"90% CI months: [{lower_90_months:.4f}, {upper_90_months:.4f}]")
print(f"90% CI years: [{lower_90_years:.4f}, {upper_90_years:.4f}]")
print()
print(f"95% CI months: [{lower_95_months:.4f}, {upper_95_months:.4f}]")
print(f"95% CI years: [{lower_95_years:.4f}, {upper_95_years:.4f}]")
print()
print(f"99% CI months: [{lower_99_months:.4f}, {upper_99_months:.4f}]")
print(f"99% CI years: [{lower_99_years:.4f}, {upper_99_years:.4f}]")
```

```
90% CI months: [125.5281, 132.6605]
90% CI years: [10.4607, 11.0550]

95% CI months: [124.8452, 133.3434]
95% CI years: [10.4038, 11.1120]

99% CI months: [123.5097, 134.6788]
99% CI years: [10.2925, 11.2232]
```

Testing Hypotheses

In [18]:
```
z = (estimate - 120)/standard_error_months
p = 2 * norm.cdf(-np.abs(z))

z_years = (estimate_years - 10)/standard_error_years
p_years = 2 * norm.cdf(-np.abs(z_years))

print(f"z-statistic: {z:.4f}, p-value: {p}")
print(f"z-statistic: {z_years:.4f}, p-value: {p_years}")
```

```
z-statistic: 4.1950, p-value: 2.729301504361327e-05
z-statistic: 4.1950, p-value: 2.7293015043612702e-05
```

With the calculated z-statistic, we get a p-value well below all three significance levels of $\alpha = 0.10, 0.05, 0.01$ for both months and years. This means at all three significance levels we reject the null hypothesis that $\mu = 120$ for months and 10 for years.

This decision is confirmed by our confidence intervals. 120 months and 10 years both do not fall in any of their three corresponding confidence intervals, meaning these intervals also tell us that we reject the null hypothesis at $\alpha = 0.10, 0.05, 0.01$.

Interpretation

Our estimate approximates that the mean time to death, including both living and deceased patients, is about 129 months or about 11 years with about 2.16791 and 0.18066 standard error respectively. Our confidence intervals tell us that we cannot reject mean values from $[125.5281, 132.6605]$ for months and $[10.4607, 11.0550]$ for years at the $\alpha = 0.10$ level, values from $[124.8452, 133.3434]$ for months and $[10.4038, 11.1120]$ for years at the $\alpha = 0.05$ level, and values from $[123.5097, 134.6788]$ for months and $[10.2925, 11.2232]$ for years at the $\alpha = 0.01$ level. If we calculated these intervals many times, approximately 90%/95%/99% of them would contain fixed, unknown mean time to death (months or years). Our test hypothesis test tells us that we reject the null hypothesis and conclude that the mean time to death is not equal to 120 months or 10 years.

Since this data is right-censored, we have not observed the death of some of the patients. We cannot use this test to estimate unrestricted mean lifetime because not all of our data is the definitive time of death for some patients, thus cannot be representative of the population.

Question 7

For `./data/nhanes_data_17_18.csv`, use `AgeInYearsAtScreening` to select respondents aged 18 or older with observed `SleepHoursWeekdaysOrWorkdays`. Estimate mean weekday sleep and test $H_0 : \mu = 8$ hours. Comment on whether statistical and practical significance tell the same story. For this exercise, treat observations as independent and unweighted, ignoring the complex survey design. Do not interpret these results as nationally representative population estimates or survey-adjusted confidence intervals.

In [19]:
```
nhanes = pd.read_csv('nhanes_data_17_18.csv')
older18 = nhanes.loc[nhanes['AgeInYearsAtScreening'] >= 18]
sleep = older18['SleepHoursWeekdaysOrWorkdays']
sleep = sleep.dropna()

estimate = sleep.mean()
print(f"Mean weekday sleep hours = {estimate:.4f}")
```

```
Mean weekday sleep hours = 7.6212
```

```
C:\Users\lchop\AppData\Local\Temp\ipykernel_31292\2988439396.py:1: DtypeWarning: Col
umns (142) have mixed types. Specify dtype option on import or set low_memory=False.
  nhanes = pd.read_csv('nhanes_data_17_18.csv')
```

We estimate that the average weekday sleep hours for respondents 18 or older is 7.6212.

Bootstrapping

In [20]:
```
## Bootstrapping:
R = 2_000

np.random.seed(5012)    # Setting the random seed

sample_means = []

for t in range(R):
    # Generating the sample
    idx = np.random.randint(0, len(sleep), len(sleep))
    sample = sleep.iloc[idx]

    # Proportion of 'Yes'
    o_mean = sample.mean()

    sample_means.append(o_mean)
```

Estimating $SE(\hat{\theta})$

In [21]:
```
standard_error = np.std(sample_means, ddof=1)
print(f'standard error = {standard_error:.5f}')
```

```
standard error = 0.02255
```

Constructing 90%, 95%, 99% CIs

In [22]:
```
# For 90%
lower_90 = estimate - 1.645 * standard_error
upper_90 = estimate + 1.645 * standard_error

# For 95%
lower_95 = estimate - 1.960 * standard_error
upper_95 = estimate + 1.960 * standard_error

# For 99%
lower_99 = estimate - 2.576 * standard_error
upper_99 = estimate + 2.576 * standard_error

print(f"90% CI: [{lower_90:.4f}, {upper_90:.4f}]")
print(f"95% CI: [{lower_95:.4f}, {upper_95:.4f}]")
print(f"99% CI: [{lower_99:.4f}, {upper_99:.4f}]")
```

```
90% CI: [7.5842, 7.6583]
95% CI: [7.5770, 7.6654]
99% CI: [7.5632, 7.6793]
```

Test Hypotheses

In [23]:
```
z = (estimate - 8)/standard_error
p = 2 * norm.cdf(-np.abs(z))

print(f"z-statistic: {z:.4f}, p-value: {p}")
```

```
z-statistic: -16.7975, p-value: 2.546310450309766e-63
```

With the calculated z-statistic, we get a p-value below all three significance levels of $\alpha = 0.10, 0.05, 0.01$. This means at all three significance levels we reject the null hypothesis that $\mu = 8$ hours.

This decision is confirmed by our confidence intervals. 8 does not fall in any of the three confidence intervals, meaning these intervals also tell us that we reject the null hypothesis at $\alpha = 0.10, 0.05, 0.01$.

Interpretation

Our estimate approximates that the average weekday sleep hours for respondents 18 or older is about 7.6212 with about 0.02255 standard error. Our confidence intervals tell us that we cannot reject mean values from $[7.5842, 7.6583]$ at the $\alpha = 0.10$ level, values from $[7.5770, 7.6654]$ at the $\alpha = 0.05$ level, and values from $[7.5632, 7.6793]$ at the $\alpha = 0.01$ level. If we calculated these intervals many times, approximately 90%/95%/99% of them would contain fixed, unknown mean sleep hours of respondents 18 or older. Our test hypothesis test tells us that we reject the null hypothesis and conclude that the average weekday sleep hours is not equal to 8.

Statistical significance is a reliability claim. It says that if we conduct many, many trials, we observe something to be the case. Practical significance is more concrete. It is not so much an observation as it is a fact that you can claim.